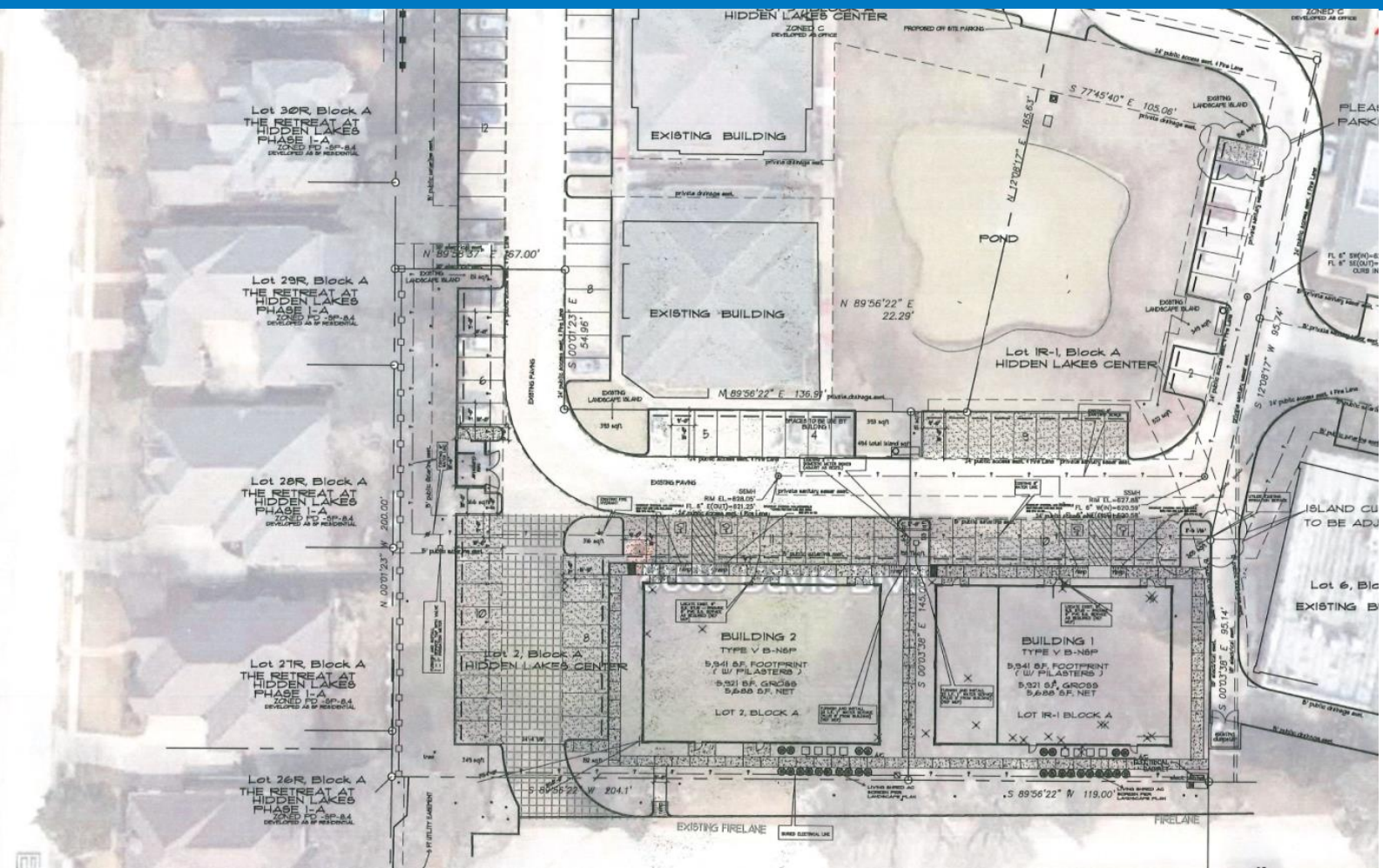
esri Malaysia

11 FEBRUARY 2026

E-Plans Submission System Phase 1

User Requirement Study Report

v1.2

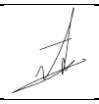


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1.1	2.1.2	The submission process workflow has been revised, removing both the "Download Processing Output" and the "Data Enhancement Step".	04/02/2026
1.1	2.3.2.1	The mock-up of the Admin Listing has been updated with a new header background.	04/02/2026
1.1	2.3.2.2	The mock-up of the PSP User Listing has been updated with a new header background.	04/02/2026

1.1	2.4.2	The mock-up of the Profile Settings has been updated with a new header background.	04/02/2026
1.1	2.5.1	The "Download Processing Output" and the "Data Enhancement Step" are removed from the use case diagram.	04/02/2026
1.1	2.5.2.1	The mock-up of the Plan Submission Listing has been updated with a new header background. The "Atribut Diperlukan" and "Sedia Untuk Dihantar" statuses have been removed and replaced with the "Draf" status.	04/02/2026
1.1	2.5.2.2	The mock-up of the New Submission page has been updated with a new header background. Excel has been added as a required field for initiating a new submission.	04/02/2026
1.1		The Data Enhancement Step (previous section 4.5.2.3) has been removed.	04/02/2026
1.1	2.5.2.3	The mock-up of the Plan Submission Summary page has been updated with a new header background. The ETL conversion process (previously handled within section 4.5.2.2) has been moved to / is now handled within the Submit for Admin Approval stage.	04/02/2026
1.1	2.6.2.1	The mock-up of the Plan Approval Listing has been updated with a new header background.	04/02/2026
1.1	2.6.2.2	The mock-up of the Admin Approval page has been updated with a new header background.	04/02/2026
1.1	3.2.2	Sample of CAD Template Layer Naming has been updated with ID layers.	04/02/2026

1.1	3.3	Guidelines of Excel File Preparation. This new section explains the preparation of Excel Template to ensure seamless integration between CAD drawings and the GIS database.	04/02/2026
1.2	3.4.4	Change the wording MBSA to MBSP.	11/02/2026
1.2	3.5.1	Change the value for tajuk (Field Name) length from 4000 to 10000.	11/02/2026

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LIST OF ABBREVIATIONS

Table 1: List of Abbreviation

ABBREVIATION	FULL NAME
GIS	Geographical Information System
MBSP	Majlis Bandaraya Seberang Perai
PSP	Pesticide Submissions Portal
URL	Uniform Resource Locator
URS	User Requirement Study
VM	Virtual Machine

***

INTRODUCTION

1.1 Purpose

This User Requirement Specification document describes the business and functional requirements for Majlis Bandaraya Seberang Perai (MBSP)'s E-Plan Submission system. The requirement includes the business needs, specifications, requirements, input, output, functions and workflow. All the listed designs, and only these, will be developed according to their priorities. Any variation to this design requires a raise of a Change Request. This document will be reviewed and updated accordingly throughout the implementation of the project.

1.2 Scope of Work

This document adheres to the Scope of Work documentation as submitted Majlis Bandaraya Seberang Perai (MBSP).

Table 2: Scope of Work

NO	SCOPE OF WORK	ACTIVITIES
1	User Requirement Study (URS)	1. E-Plan web system (admin & user). 2. Custom form plan / data submission. 3. E-Plan based data. 4. ETL Workbench. 5. User Requirement Study report.
2	Installation and Configuration	1. ArcGIS Data Interoperability Server. 2. ETL Workbench.
3	Module Development	1. E-Plan web system (admin). 2. E-Plan web system (user). 3. Custom form plan / data submission.

		4. Notifications and status update.
4	User Acceptance Test	1. Test script preparation. 2. User Acceptance Test (UAT). 3. Final Acceptance Test (FAT). 4. User Acceptance Test report.
5	Training and Post Implementation	1. Web Application (2 days x 10 pax). 2. ETL Workbench (3 days x10 pax). 3. Post Implementation period for 3 months.

1.3 Project Implementation

The project will be implemented in several stages as shown in Figure 1 below.

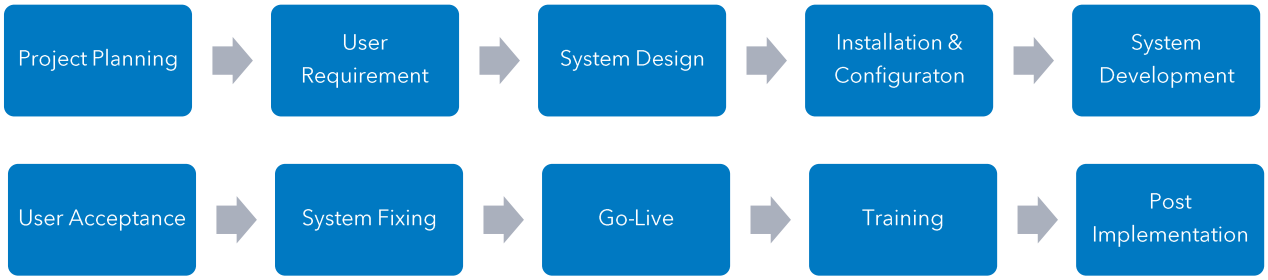


Figure 1: Project Implementation Stages

E-PLAN SUBMISSION SYSTEM

2.1 Model Requirements

The primary objective of the E-Plan Submission system is to create a comprehensive digital platform that modernizes and streamlines the plan submission process for MBSP. By eliminating the inefficiencies of manual processes, this solution expedites the plan submission application process and delivers a seamless, automated workflow for both developers and MBSP.

The E-Plan Submission system provides PSP users with a user-friendly web registration platform, enabling them to submit plans online using standardized templates and removing the need for physical interactions with MBSP's internal officers. The E-Plan Submission system empowers the *Jabatan Perancang Bandar* to efficiently review and process the development plans submitted by PSP users.

2.1.1 Overall System Access

The user creation process for the E-Plan Submission system is handled externally through the ArcGIS Portal, where the Portal Admin bears the primary responsibility for user management. All biographical information for users in the E-Plan Submission system is synchronized directly from the ArcGIS Portal. No additional information is collected or required.

The E-Plan Submission system depicts two entities, MBSP and PSP:

1. Under MBSP, there are two user roles:

- Super Admin (refer to 2.2.1 Super Admin Module).
- Admin (refer to 2.2.2. Admin Module).

2. Under PSP, there is one user role:

- PSP User (refer to 2.2.3. User Module).



Figure 2: Mockup Landing Page

2.1.2 Overall Submission Process

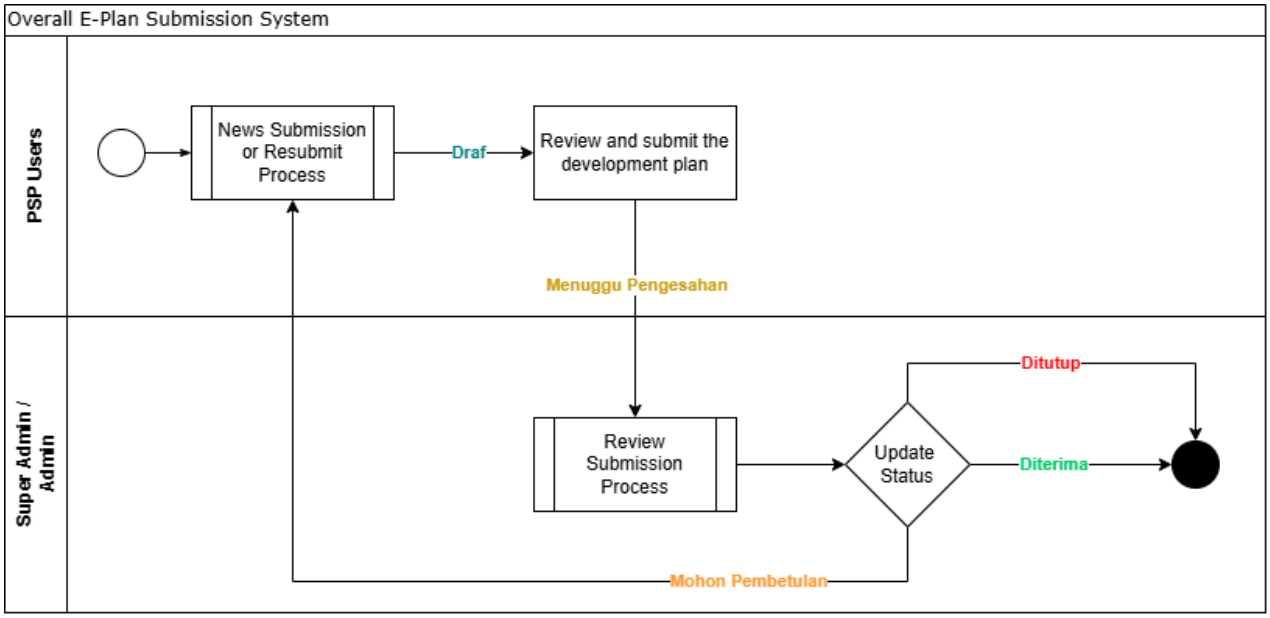


Figure 3: Overall E-Plan Submission Process

As illustrated in Figure above, the overall plan submission workflow in the E-Plan Submission system is structured as follows:

1. **Initial Submission**

   The process begins with the PSP user initiating a submission by uploading the development plan, including CAD drawing files and excel, along with the required application details. Upon successful submission, the plan is automatically assigned the status ***Draf*** (Draft).

2. **Review and Submission for Approval**

   The PSP user conducts a final review on the detailed summary page and submits the development plan for admin approval. Upon successful submission, the status is updated to ***Menunggu Pengesahan*** (Awaiting Approval).

3. **Admin Review and Decision**

   Upon reviewing the submitted plan and conducting quality control on the final CAD-to-GIS results, the Admin can update the submission status to one of the following:

   - Diterima (Verified).
   - Ditutup (Discontinued).
   - Mohon Pembetulan (Request Revision).

2.2 Role-Based Access Control

The E-Plan Submission system distinguishes between three specific user roles, each with distinct access rights to various system modules:

1. **Super Admin**

   Super Admin possesses the highest level of access, encompassing comprehensive administrative controls and system monitoring capabilities that significantly influence the overall functionality of the E-Plan Submission System.

2. **Admin**

   Admins handle day-to-day management tasks within the system, which include reviewing, approving, and publishing approved submissions.

3. **PSP User**

   Non-MBSP or public users are limited to managing their own submissions and personal profile settings within the system.

2.2.1 Super Admin Module

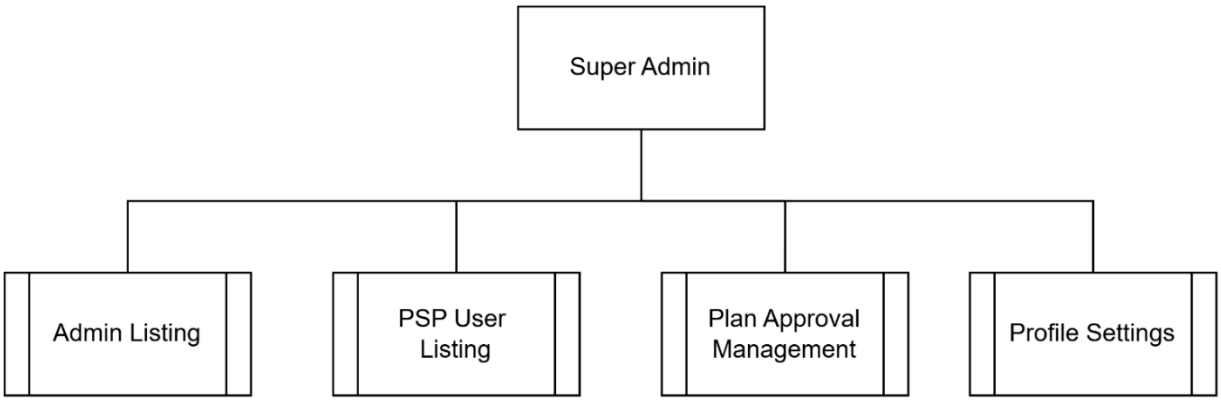


Figure 4: Super Admin Module Access

This section outlines the modules available to the Super Admin, who can navigate through all modules, demonstrating their extensive control over the system. Below is a list of modules for the Super Admin, with references to sections in this document:

1. Admin Listing (refer to 2.3.1 Admin Listing).
2. PSP User Listing (refer to 2.3.2 PSP User Listing).
3. Plan Approval Management (refer to 2.6 Plan Approval Management).
4. Profile Settings (refer to 2.4 Profile Settings).

2.2.2 Admin Module

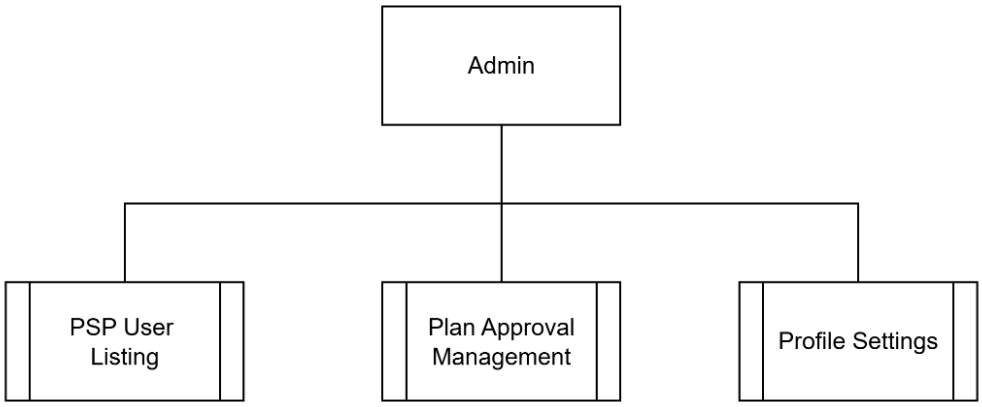


Figure 5: Admin Module Access

This section describes the modules available to Admin users. Unlike the Super Admin, Admins have slightly restricted access to maintain operational integrity and security. Specifically, the PSP User Listing module is excluded from their access. Below is a list of modules available to Admin users, with references to sections in this document:

1. PSP User Listing (refer to 2.3.2 PSP User Listing)
2. Plan Approval Management (refer to 2.6 Plan Approval Management)
3. Profile Settings (refer to 2.4 Profile Settings)

2.2.3 PSP User Module

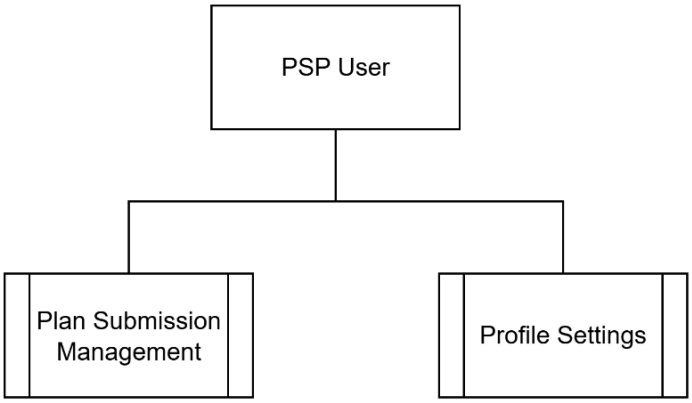


Figure 6: PSP User Module Access

This section outlines the modules accessible to PSP users. These users focus primarily on personal application tasks, restricted only to plan submission management and profile customization to ensure user independence and data privacy. Below is a list of modules available to PSP users, with references to sections in this document:

1. Plan Submission Management (refer to 2.5 Plan Submission Management).
2. Profile Settings (refer to 2.4 Profile Settings).

2.3 User Listing

2.3.1 Use Case Diagram

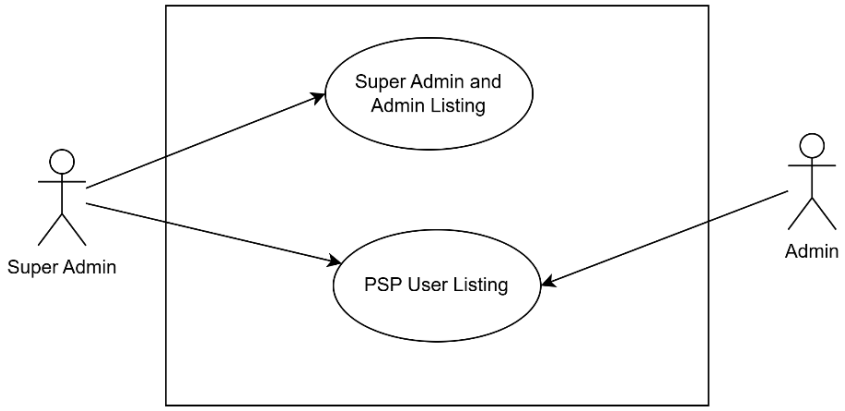


Figure 7: User Listing Use Case Diagram

Table below explain the details of the use case diagram.

Table 3: User Case Diagram Explanation for User Listing Module

NO	USE CASE	SUMMARY
1	Super Admin and Admin Account Listing	Super Admin will be able to see all the registered Super Admin and Admin list. Super Admin can search, select and view the details.
2	PSP User Listing	Super Admin and Admin both can view all PSP user account registered in the system.

2.3.2 System Perspective

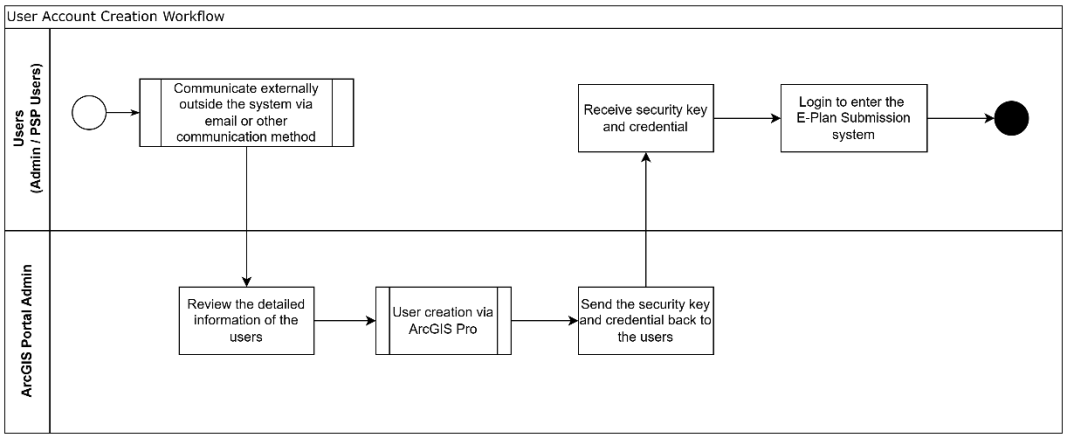


Figure 8: New User Account Workflow

User management is primarily handled through the ArcGIS Portal, with user creation managed externally. This process is initiated through external communication (such as email or other methods) between the users and MBSP. Overall user creation process above will the same for both admin creation and PSP user creation. Following this, MBSP extends an invitation to the user and provides the necessary credentials to access the E-Plan Submission system.

2.3.2.1 Admin Listing

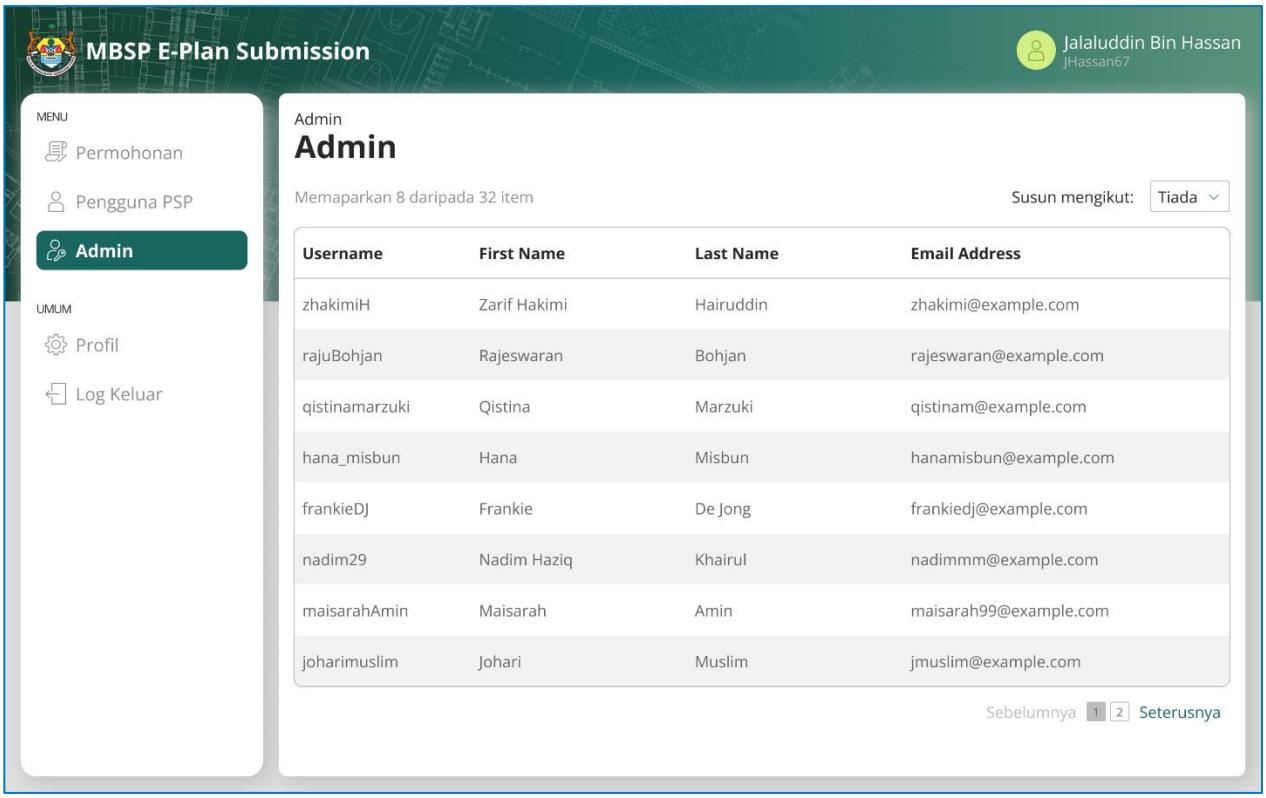


Figure 9: Admin User Listing

The E-Plan Submission system provides a detailed listing of all registered Admin accounts drawn from the ArcGIS Portal, enabling streamlined administration, proactive monitoring, and ensured compliance with established data governance standards.

2.3.2.2 PSP User Listing

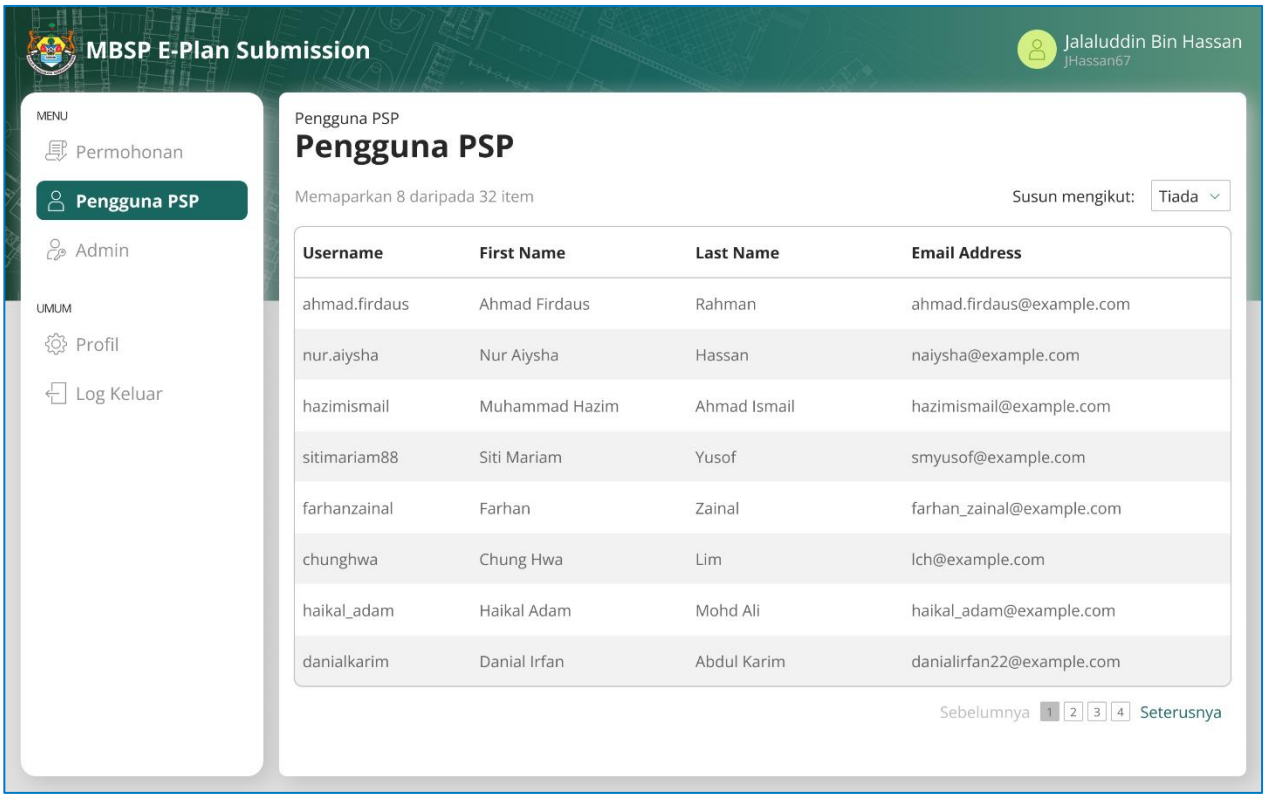


Figure 10: PSP User Listing

The E-Plan Submission system offers a comprehensive view of all registered PSP user accounts synchronized from the ArcGIS Portal. This facilitates effective oversight and management, supporting timely user access control and adherence to data governance policies.

2.4 Profile Settings

2.4.1 Use Case Diagram

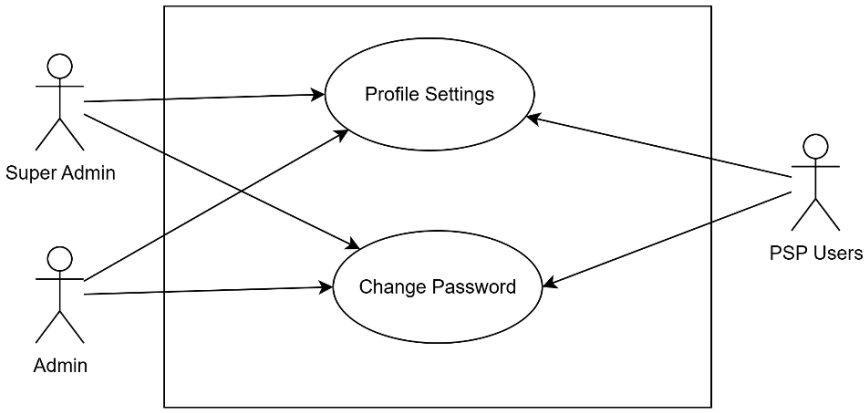


Figure 11: Profile Settings Use Case Diagram

Table below explain the details of the use case diagram.

Table 4: Use Case Diagram explanation for Profile Settings

NO	USE CASE	SUMMARY
1	Profile Settings	Every registered account will be able to view their own account details. They can update their account information at any time.
2	Change Password	Every registered account will be able to update their own password.

2.4.2 System Perspective

The E-Plan Submission system provide users to update their own basic details, email, and password within a software system is crucial for maintaining both security and personalization. This autonomy empowers users to manage their account information efficiently and securely. The Profile Settings module is accessible to both Admin and PSP users. Any changes made to the profile within the E-Plan Submission system will also be synchronized and take effect on the ArcGIS Portal.

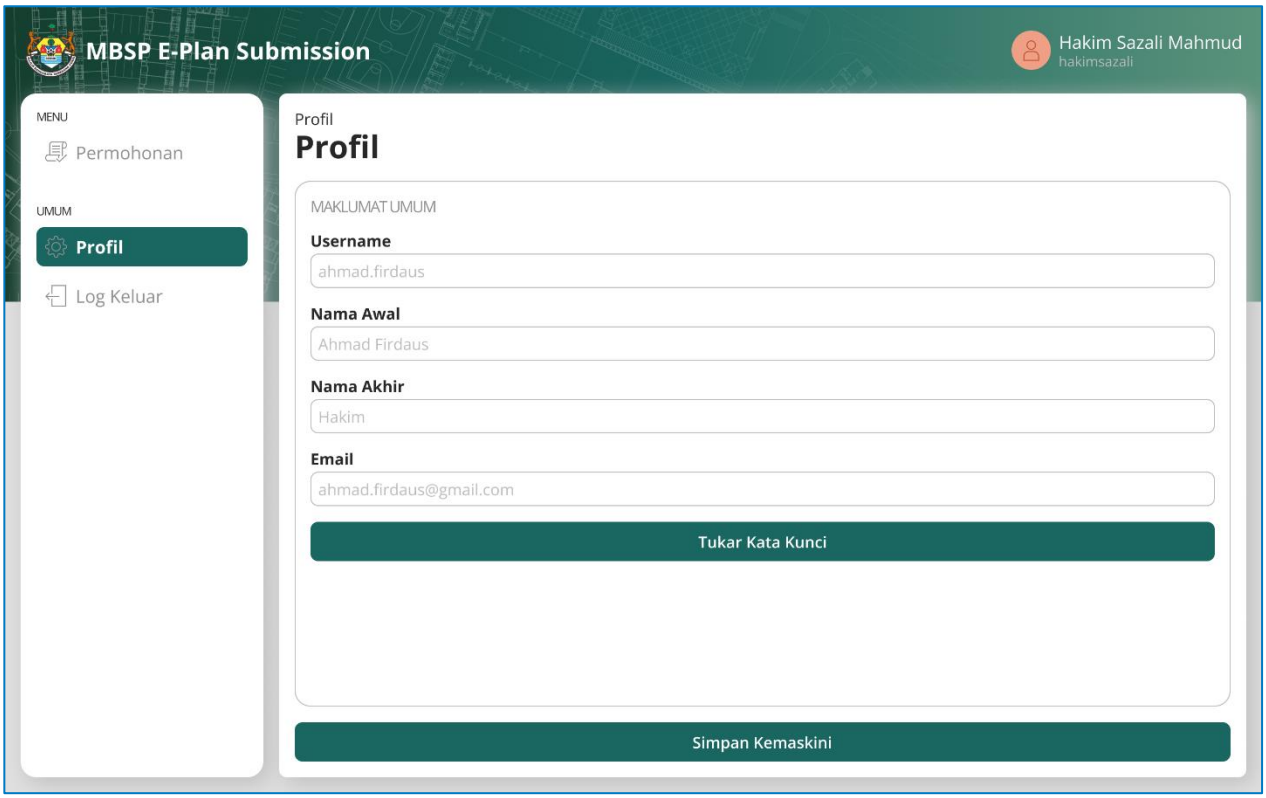


Figure 12: Profile Settings Page

Table below shows the content of the Profile Settings page.

Table 5: Profile Settings Contents

NO	FIELD	TYPE	DESCRIPTION
1	Username (Primary Key)	String	Non-editable
2	*Nama Pertama* (First Name)	String	
3	*Nama Akhir* (Last Name)	String	
4	*E-mel* (Email Address)	String	
5	*Kata Laluan* (Password)	String	Follow the ArcGIS Portal Organization's password policy set by MBSP: • Minimum 8 characters. • Contain at least one letter, one symbol and one number. • No space.

2.5 Plan Submission Management

2.5.1 Use Case Diagram

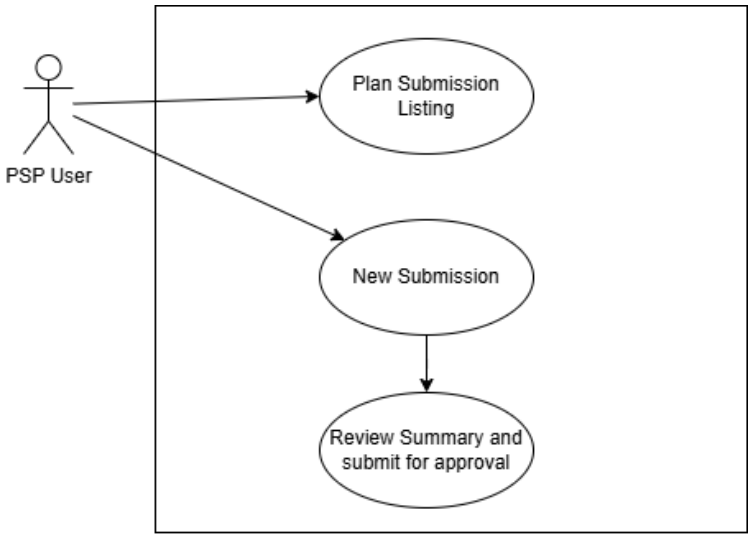


Figure 13: Plan Submission Management Use Case Diagram

Table below explain the details of the use case diagram.

Table 6: User Case Diagram Explanation for Plan Submission Management Module

NO	USE CASE	SUMMARY
1	Plan Submission Listing	PSP users can view a list of only their own submitted items. They can select any item to access detailed information and the current status of their plan submission.
2	New Submission	PSP users can perform an initial submission by uploading the development plan, including CAD drawing files and excel, along with the required application detail.
5	Plan Submission Summary	PSP users can review a summary of their plan submission prior to finalizing and sending it to the Admin for approval.

2.5.2 System Perspective

2.5.2.1 Plan Submission Listing

The system provides a comprehensive list of items submitted by the user themselves. From this list, PSP users can track real-time updates and monitor the current status of their plan development for tracking purposes and to support decision-making.

The list will display items with the following statuses:

- ***Draf*** (Pending attribute data)
- ***Menunggu Pengesahan*** (Awaiting approval)
- ***Diterima*** (Verified)
- ***Ditutup*** (Discontinued)
- ***Mohon Pembetulan*** (Request revision)

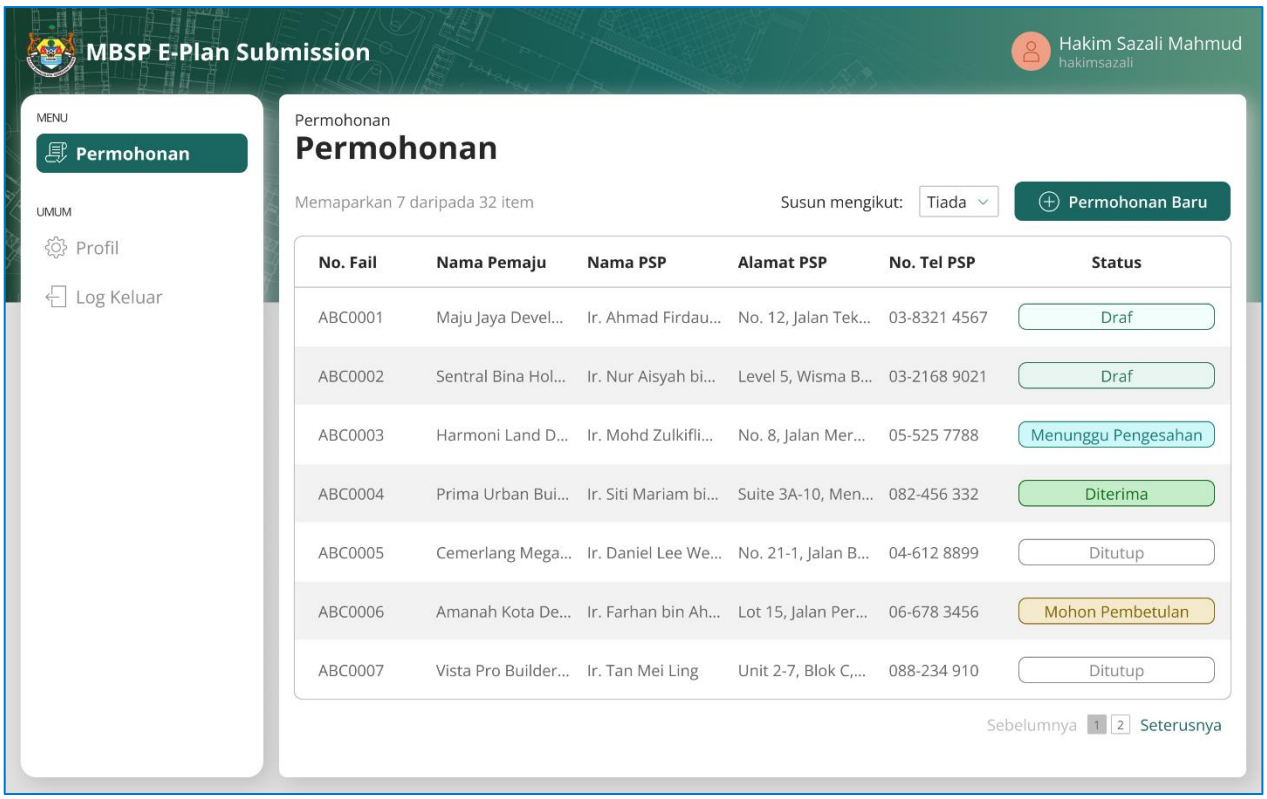


Figure 14: PSP User - Plan Submission Listing

2.5.2.2 New Submission

This page requires users to complete all mandatory fields and upload the **necessary CAD drawing files and excel** in order to initiate a new submission. Upon successful submission, a new item will be created in the listing with the status ***Draf*** (Draft).

PSP users will receive real-time status updates throughout the process, ensuring they are promptly informed of the change and can monitor progress directly from the submission listing.

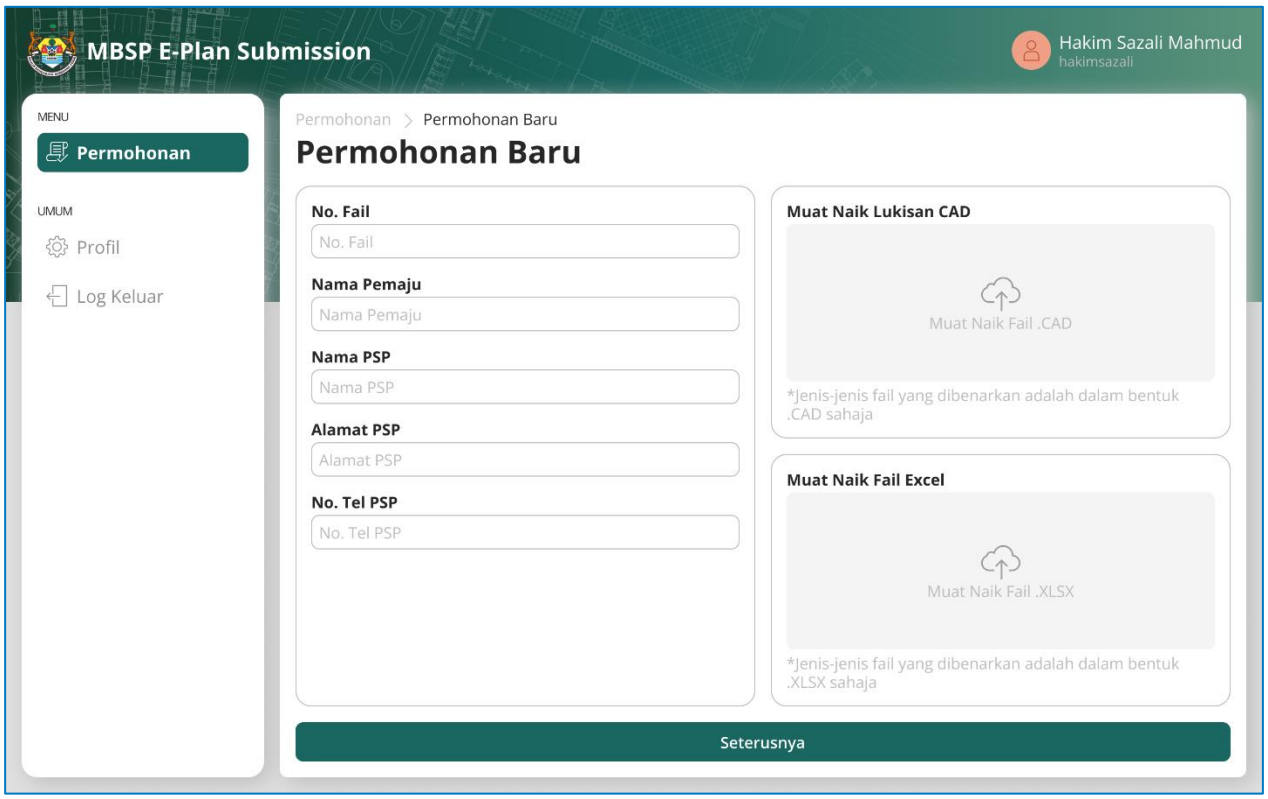


Figure 15: PSP User - New Submission

Table below shows the content of the New Submission Page.

Table 7: New Development Plan Submission Contents

NO	FIELD	TYPE	DESCRIPTION	REMARKS
1	*No fail* (Primary Key)	String	Non-editable after plan creation.	Required
2	*Nama Pemaju*	String		Required
3	*Nama PSP*	String		Required
4	*Alamat PSP*	String		Required
5	*No tel PSP*	String		Required
6	CAD drawing files	File	.dwg & .dxf.	Required
7	Excel_template	File	.xlsx	Required

2.5.2.3 Plan Submission Summary

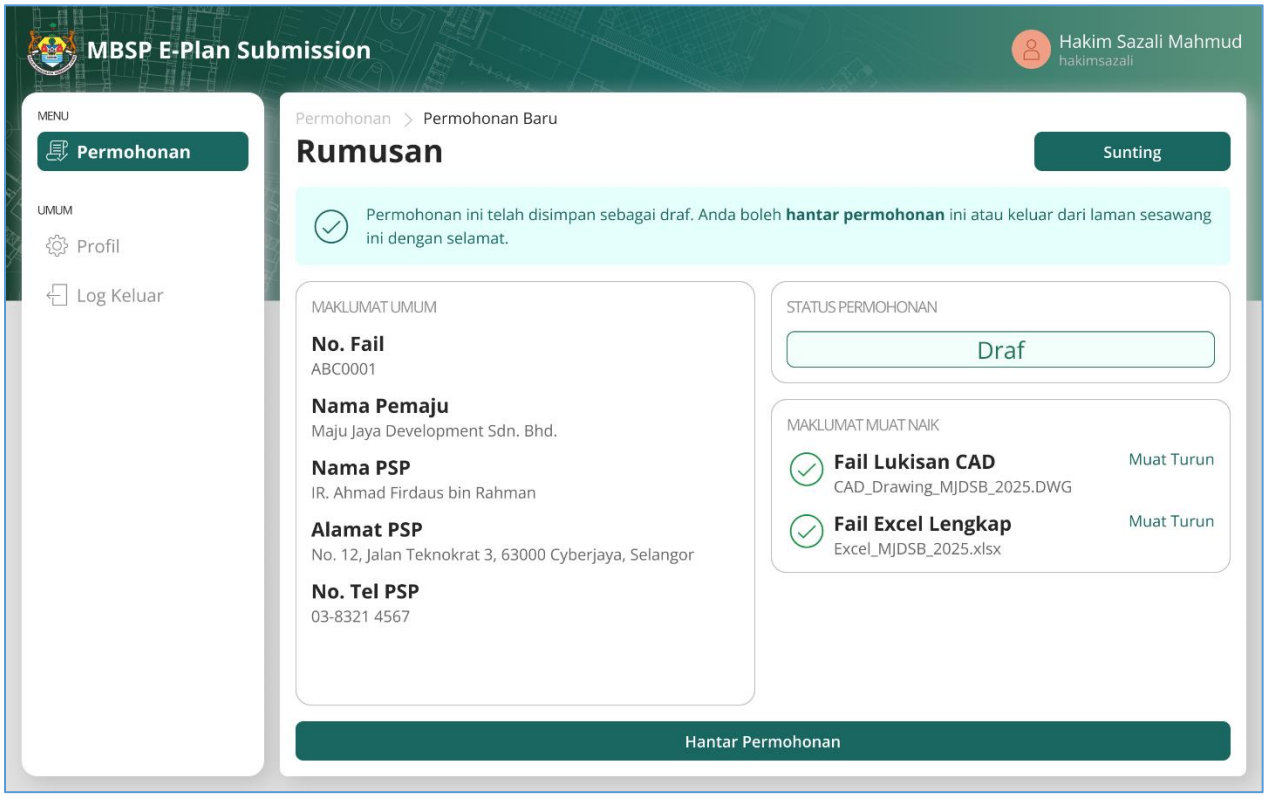


Figure 16: Summary Page Before Submitting Submission

The E-Plan Submission system includes a dedicated detailed information page that allows PSP users to thoroughly review all data, components, and details of their development plan prior to final submission.

Once the review is complete, PSP users can click the submit button to forward the application for ETL conversion. If any ETL conversion process fails, the system will notify the user of the error, and the user will be required to correct the issue and upload the proper CAD drawing files and excel.

Upon successful submission, a success message will be displayed, and the application's status will be updated to ***Menunggu Pengesahan*** (Awaiting Approval), pending admin approval.

Table below shows the content of the PSP User' Plan Submission Summary Page.

Table 8: PSP User Plan Submission Content

NO	FIELD	TYPE	DESCRIPTION
1	*No fail*	String	Readable only
2	*Nama Pemaju*	String	Readable only
3	*Nama PSP*	String	Readable only
4	*Alamat PSP*	String	Readable only
5	*No tel PSP*	String	Readable only
6	CAD drawing files	File	Readable only
7	Map Image	File	Readable only
8	Excel_template	File	Readable only

2.6 Plan Approval Management

2.6.1 Use Case Diagram

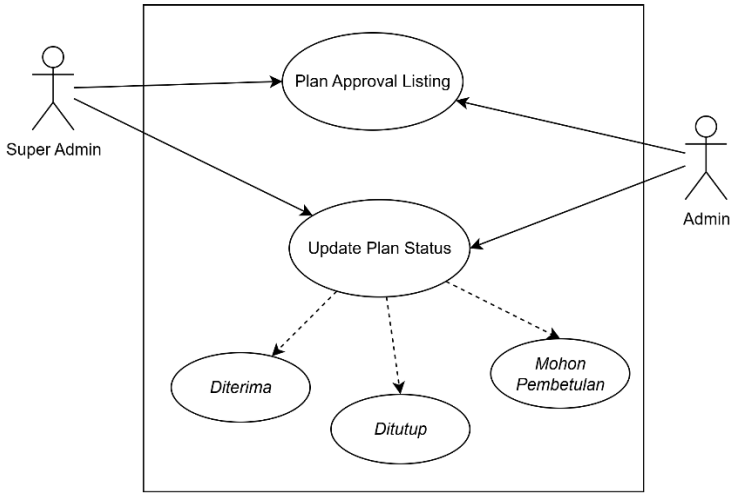


Figure 17: Plan Submission Approval Use Case Diagram

Table below explain the details of the use case diagram.

Table 9: User Case Diagram Explanation for Plan Submission Approval Module

NO	USE CASE	SUMMARY
1	Plan Approval Listing	Super Admins and Admins can view a comprehensive list of all plan submission applications submitted by every PSP user.
2	Update Plan Status	Super Admins and Admins can update the status of submissions to one of the following status: • *Diterima* (Verified) • *Ditutup* (Discontinued) • *Mohon Pembetulan* (Request Revision)

2.6.2 System Perspective

2.6.2.1 Plan Approval Listing

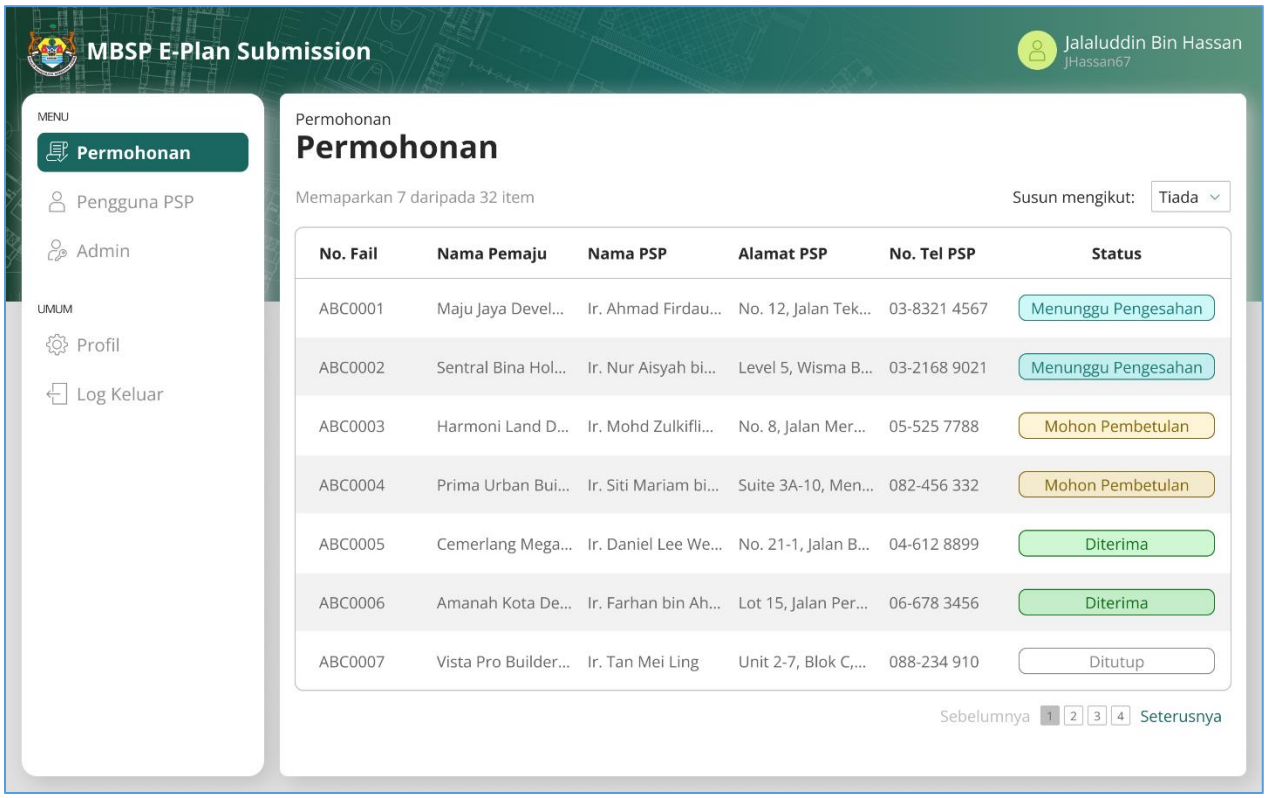


Figure 18: Admin - Plan Submission Listing

The E-Plan Submission system provides a comprehensive list of all plan submission applications submitted by every PSP user. From this listing, Admins can track real-time updates on submission statuses. This promotes greater transparency throughout the review process and enables faster, more informed decision-making.

The list will display items with the following statuses:

- ***Menunggu Pengesahan*** (Awaiting approval)
- ***Diterima*** (Verified)
- ***Ditutup*** (Discontinued)
- ***Mohon Pembetulan*** (Request revision)

2.6.2.2 Update Plan Status

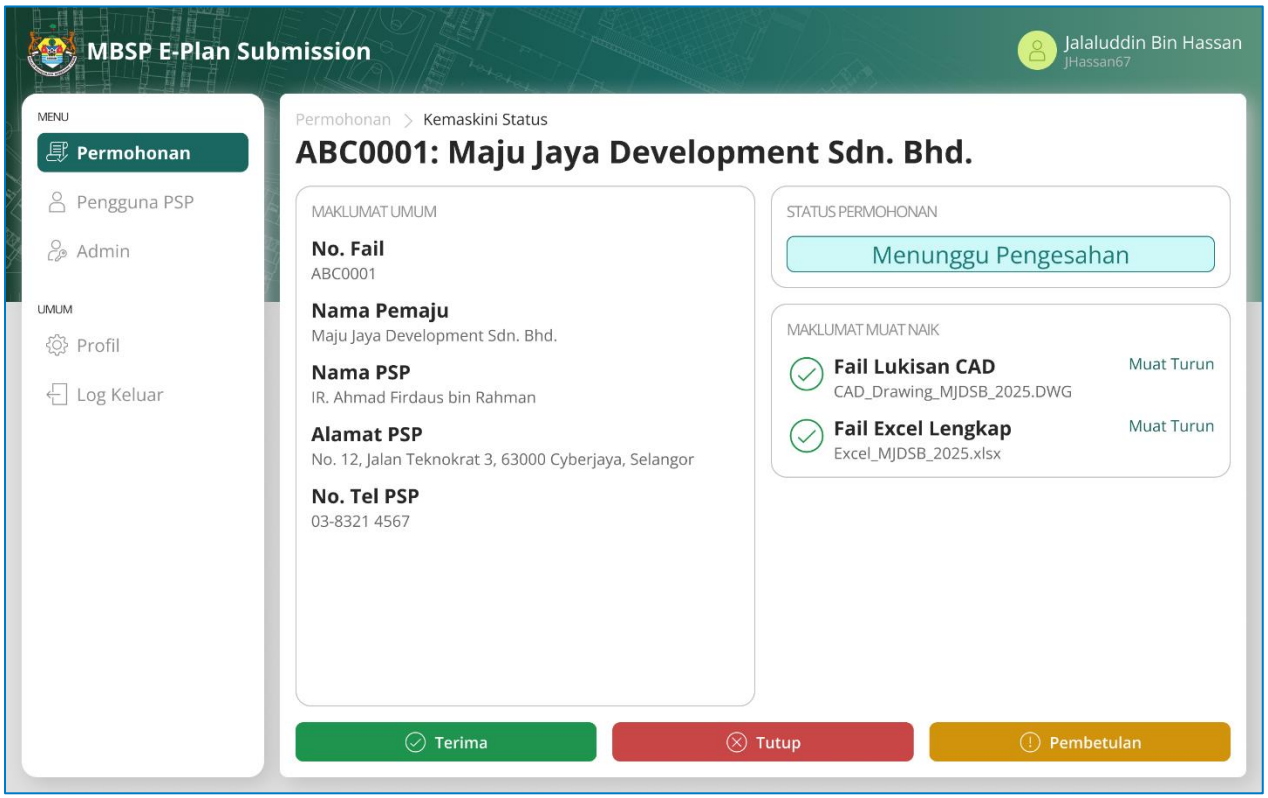


Figure 19: Admin Approval Page

The E-Plan Submission system features a dedicated detailed information page that enables Admins to comprehensively review all data and components associated with each submitted development plan.

This page includes actionable buttons allowing Admins to update the status of submissions currently in ***Menunggu Pengesahan*** *(Awaiting Approval)* by selecting one of the following options:

- ***Diterima*** (Verify the plan)
- ***Ditutup*** (Discontinue the plan)
- ***Pembetulan*** (Request revisions from the PSP user)

Upon performing any status update or related action, the system automatically provides the Admin with relevant confirmation updates and error notifications as appropriate.

Table below shows the content of the Admin Approval Page.

Table 10: Admin Approval Page Content

NO	FIELD	TYPE	DESCRIPTION
1	*No fail*	String	Readable only
2	*Nama Pemaju*	String	Readable only
3	*Nama PSP*	String	Readable only
4	*Alamat PSP*	String	Readable only
5	*No tel PSP*	String	Readable only
6	CAD drawing files	File	Readable only
7	Map Image	File	Readable only
8	Completed Excel File	File	Readable only

2.7 Supported Framework/Library Versions

This project is designed, developed, and validated specifically for the framework versions listed below. All functionalities, specifications, and implementation details described in this document are based on and aligned with these supported versions.

Use of framework versions outside the scope defined herein has not been evaluated as part of this project and may result in differences in system behavior or functionality.

Table 11: E-Plan Submission Project's List of Supported Framework/Library Versions

NO	FRAMEWORK / LIBRARY	VERSION
1	.NET (Target Framework)	10.0
2	BCrypt.Net-Next	4.0.3
3	DotNetEnv	3.1.1

NO	FRAMEWORK / LIBRARY	VERSION
4	Microsoft.AspNetCore.Authentication.JwtBearer	9.0.0
5	Microsoft.EntityFrameworkCore.Design	9.0.0
6	Swashbuckle.AspNetCore	7.2.0
7	Serilog.AspNetCore	9.0.0
8	Serilog.Sinks.Console	6.0.0
9	Serilog.Sinks.File	6.0.0
10	AspNetCoreRateLimit	5.0.0
11	AspNetCore.HealthChecks.SqlServer	9.0.0
12	AspNetCore.HealthChecks.Rabbitmq	9.0.0
13	FluentFTP	50.1.0
14	MassTransit	8.3.4
15	MassTransit.EntityFrameworkCore	8.3.4
16	MassTransit.RabbitMQ	8.3.4
17	Microsoft.CodeAnalysis.CSharp.Scripting	4.8.0
18	Microsoft.Data.SqlClient	6.1.3
19	Microsoft.Data.Sqlite	9.0.0
20	Microsoft.EntityFrameworkCore	9.0.0
21	Microsoft.EntityFrameworkCore.SqlServer	9.0.0
22	Microsoft.Extensions.Http	9.0.0
23	Microsoft.Extensions.Logging.Abstractions	9.0.4
24	MySqlConnector	2.4.0
25	Npgsql	9.0.2

NO	FRAMEWORK / LIBRARY	VERSION
26	Npgsql.EntityFrameworkCore.PostgreSQL	9.0.0
27	SSH.NET	2024.1.0
28	System.Security.Cryptography.Xml	9.0.0

FILE SUBMISSION

3.1 Existing Data

An investigative study was conducted during the User Requirement session with *Jabatan Perancang Bandar*, Majlis Bandaraya Seberang Prai (MBSP) to identify and understand the existing data that will be converted into the E-Plan Submission system. Based on the session, it was found that MBSP receives drawings from contractors in the form of softcopy DWG format. MBSP expressed the need to convert these existing plans into GIS format and store them in a centralized database. Additionally, MBSP required the system to automate checks on elements such as color codes, layouts, and layer names, which would aid MBSP in evaluating and analyzing the proposed development areas. Further details of these findings will be described in the following sub-section.

3.1.1 Format of Existing CAD Drawings

Based on the sample drawings provided by *Jabatan Perancang Bandar*, it has been identified that the CAD drawing files are in DWG format. The drawing contains information such as the drawing layers, line and polygon features, labels and colors.

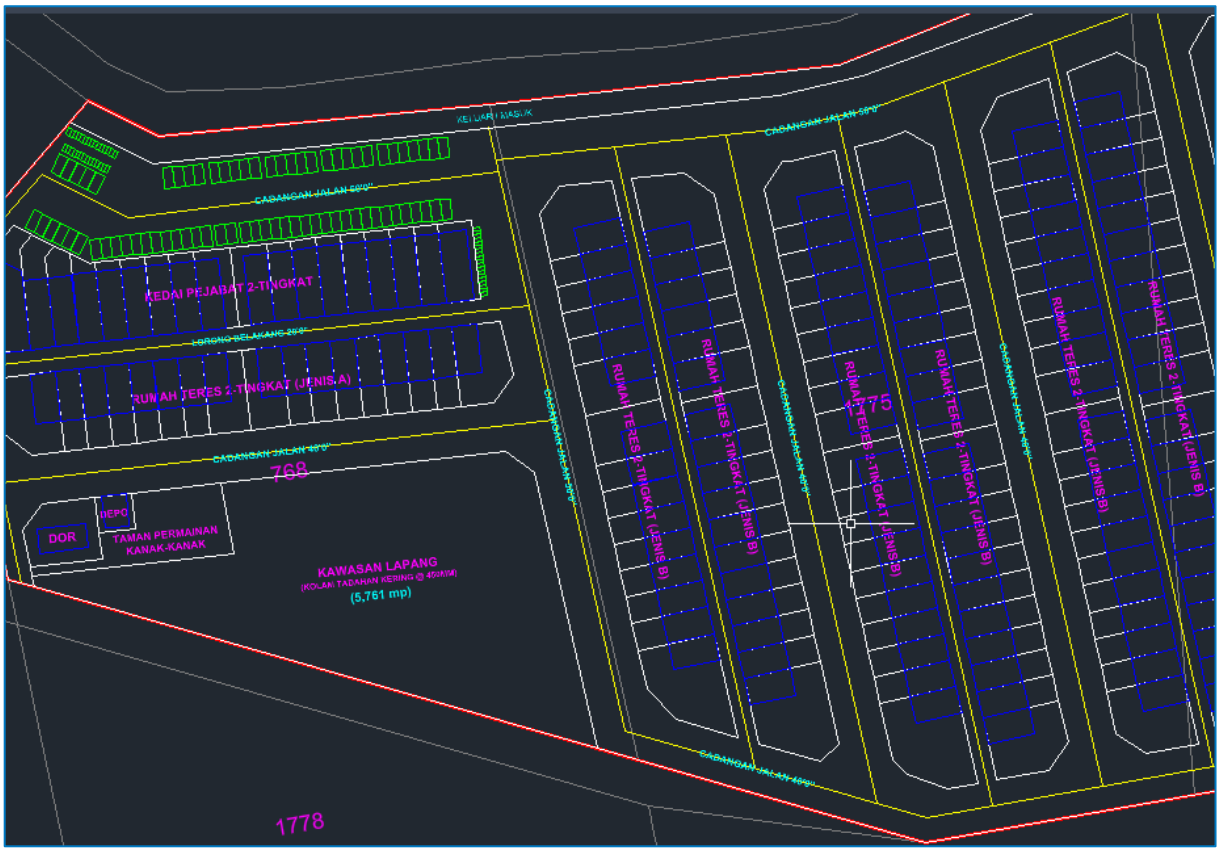


Figure 20: Example of Planning CAD Drawing

3.1.2 Layer Name Structure in CAD Drawings

Based on the CAD drawings provided by MBSP, it is evident that the layer naming is consistent throughout the drawings. The figure illustrates the layer structure based on sample drawings obtained from the *Jabatan Perancang Bandar*.

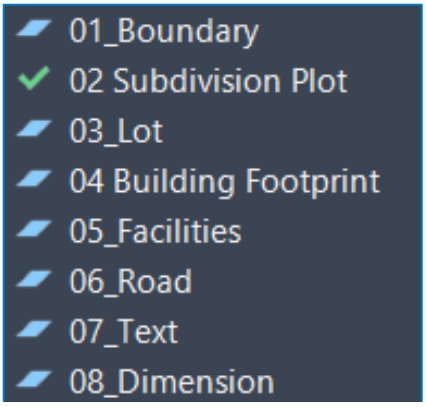


Figure 21: Layer Name Structure in Sample CAD Drawing

3.1.3 CAD Drawings Projection

GIS software interprets locations based on the defined projection. An unprojected CAD file essentially lacks a location reference. Figure below illustrates the drawing projection by overlaying the CAD drawing with the boundary layer to assess projection consistency and orientation relative to true north. Nevertheless, this method does not fully confirm the projection, as CAD software do not store coordinate system information. As a result, manual quality control in ArcGIS Desktop is necessary to validate the spatial accuracy of the converted data.

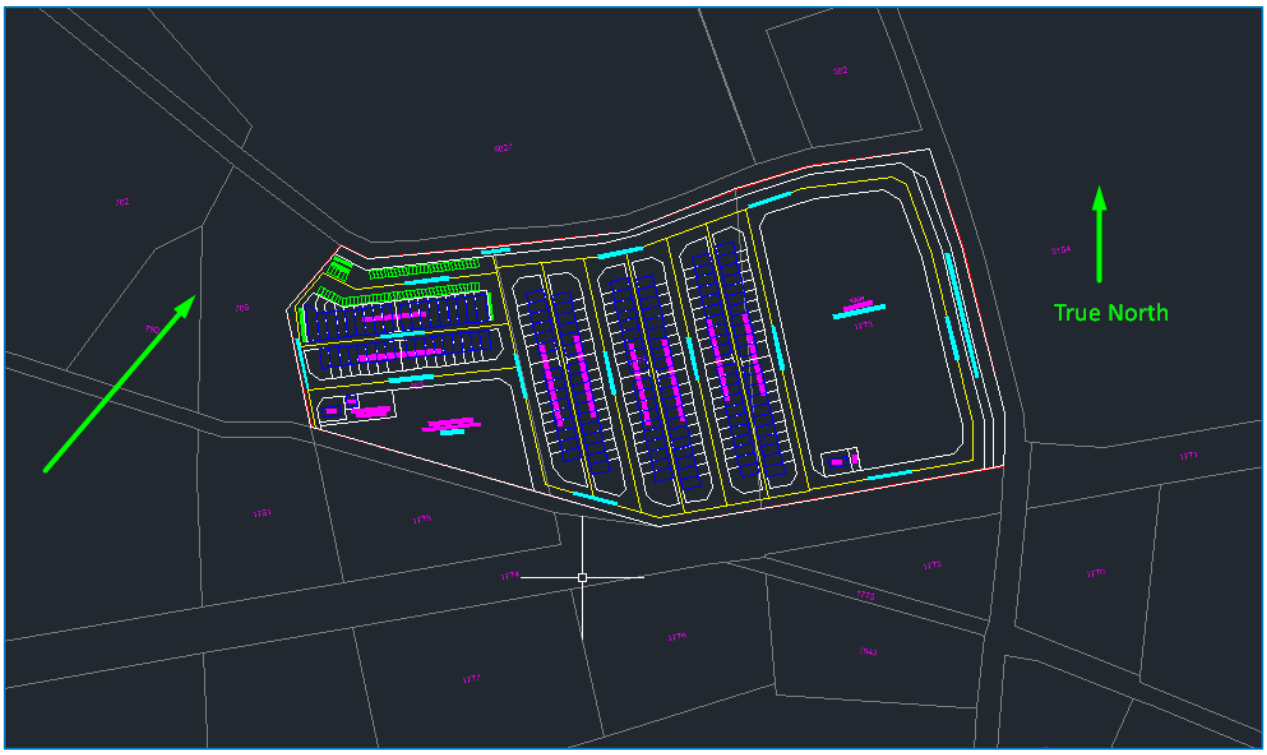


Figure 22: Green Arrow Shows Boundary Layer and True North Orientation

3.2 Guidelines of CAD Drawing Preparation

This guideline is designed to ensure consistency across all CAD drawings within a project including format, terminology and conventions. Standardization facilitates the ETL (Extract, Transform, Load) process in recognizing and interpreting data within the drawings. Elements such as layer names, symbology, and attribute definitions are standardized to minimize errors and streamline conversion. The guidelines documentation will be provided to assist in producing standard CAD files for E-Plan Submission, enhancing clarity and adherence to specifications.

3.2.1 Guidelines Workflow

This section is purposely to ensure consistency among users of the templates, thereby minimizing inconsistencies in the final CAD drawings and reducing the risk of errors during the ETL conversion to GIS data. The following are the step-by-step instructions for using the CAD Template.

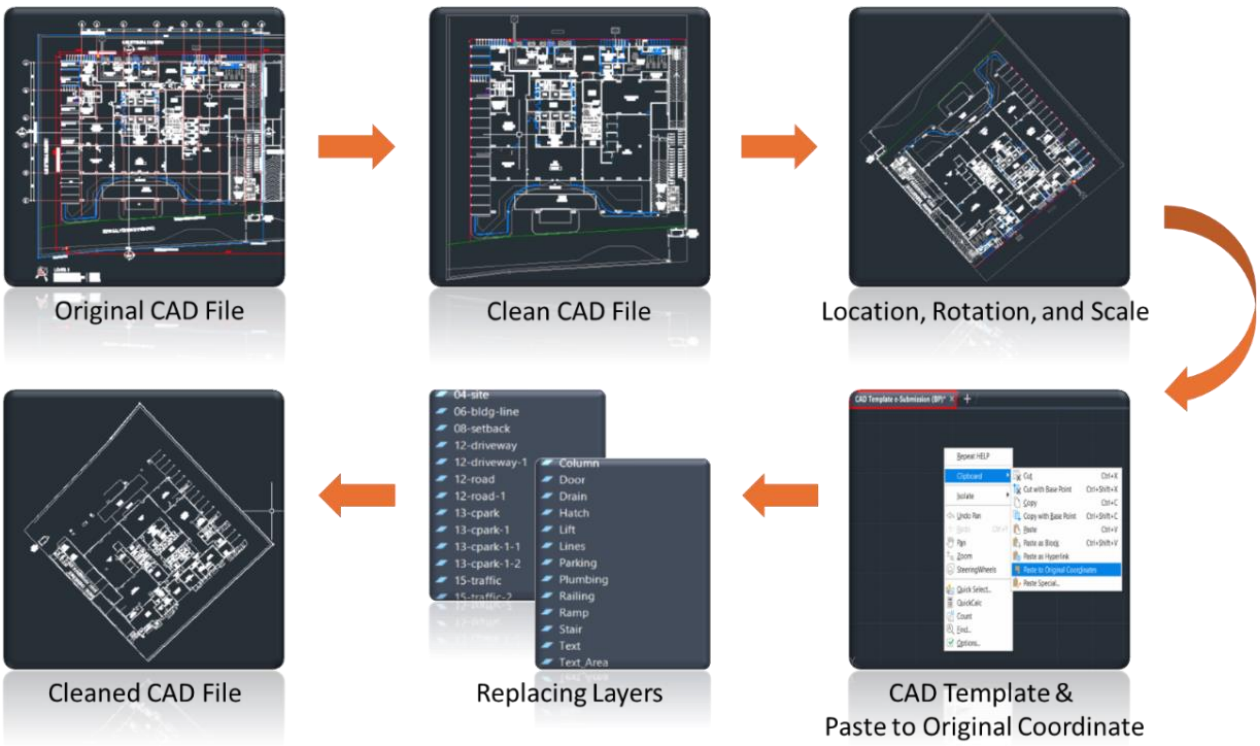


Figure 23: Guidelines Workflow Illustration

1. **Open CAD Files**: Identify CAD file for submission and open the file using CAD software.
2. **Clean CAD File**: Clean the CAD file to be converted before paste to the CAD template. Make sure to keep the features in CAD files (e.g., Plot, Footprint, & Road).
3. **Check Location, Rotation and Scale:** Make sure to check the CAD location, rotation and scale before pasting to the template. It is recommended to refer any related drawing that contains location and coordinate information. (e.g., Boundary Plan, Pre-comp Plan). Do amendments to the CAD file after checking.
4. **Download and Open CAD Template:** Download CAD template from system and open the file.
5. **Paste to Original Coordinate:** Copy the cleaned CAD features and corrected location and paste to the original coordinate in the CAD template.
6. **Replacing Layers:** The template includes standard layer names that need to be followed. Simply replace features based on the standard.
7. **Save CAD:** Save the cleaned CAD file and rename it to a new name.

3.2.2 CAD Template

CAD templates serve as a crucial link between the design world (CAD) and geographic information (GIS), establishing a standardized framework for data organization. This common language enhances the efficiency, accuracy, and reliability of the ETL (Extract, Transform, Load) process. Utilizing CAD templates can aid in identifying and automating data cleaning steps, ensuring high-quality output data. Specific formatting rules within the CAD template are designed to prevent inconsistencies or errors from entering the GIS system. Figure below illustrates the proposed layer name structure in the CAD template.



Figure 24: Proposed Layer Name Structure in CAD Template

3.3 Guidelines of Excel File Preparation

This section defines the mandatory standards for data preparation in Excel Template to ensure seamless integration between CAD drawings and the GIS database via the ETL (Extract, Transform, Load) process.

3.3.1 Guidelines Workflow

The workflow ensures that spatial data (CAD) and attribute data (Excel) are perfectly synced using a Unique Identifier (ID). The following are the step-by-step instructions for using the Excel Template.

1. **Preparation:** Prepares the plan in CAD according to the specified layer naming convention.
2. **Download and Open Excel Template:** Download Excel template from system and open the file.
3. **ID Tagging:** For every feature drawn in CAD (e.g., a specific plot), a corresponding ID must be assigned. This same ID must be entered into the respective sheet in the Excel template.
4. **Data Entry:** Populates the Excel sheets (Plot, Footprint, Road) with the necessary project metadata (e.g., *No_Pemaju*, *No_lot*).
5. **Validation & Upload:** Uploads both files to the E-Plan Submission system.

3.3.2 Excel Template

The Excel file serves as the primary attribute source. To prevent conversion errors, the following structure is mandatory:

1. Sheet Organization

The workbook must contain exactly three sheets, named as follows:

1. Plot
2. Footprint
3. Road

2. Data Mapping & ID Linking

Each sheet must include a primary key field to link with the CAD layers:

Table 12: CAD and Excel Integration Mapping

NO	EXCEL SHEET	EXCEL FIELD NAME	CAD LAYER NAME	PURPOSE
1	Plot	ID	ID_Plot	Links land plot polygon to plot data.
2	Footprint	ID	ID_Footprint	Links building polygon to footprint data.
3	Road	ID	ID_Road	Links road centerlines to road data.

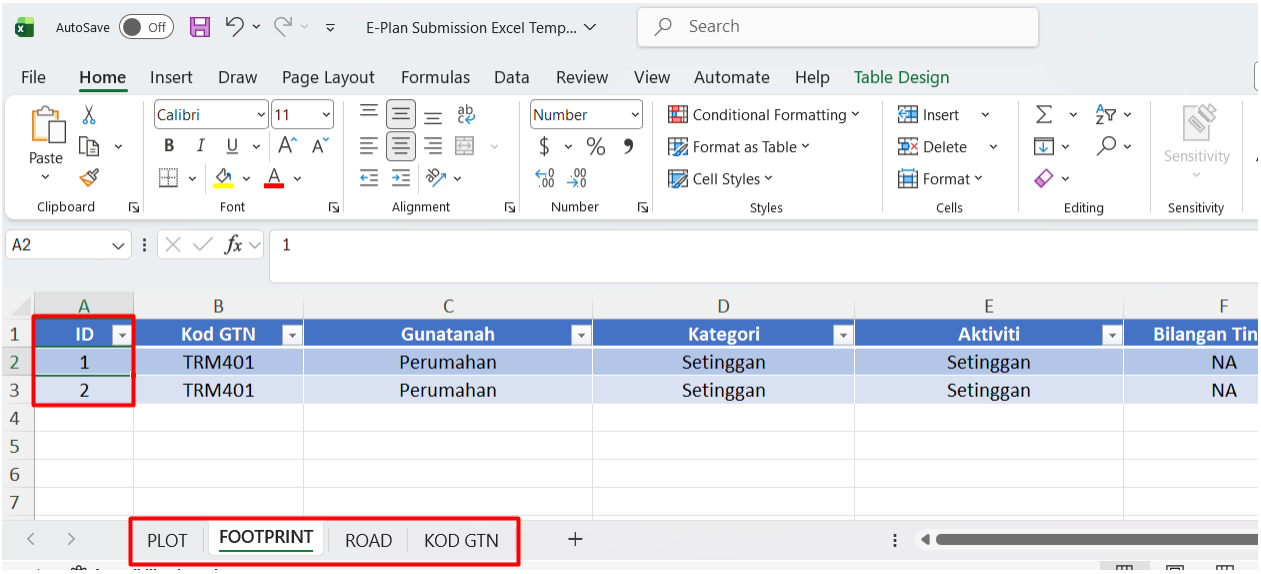


Figure 25: Proposed Excel Template

3.4 CAD File Conversion to GIS

MBSP described a need to convert drawings into GIS data format for visualization. A comprehensive data conversion methodology will be developed to seamlessly convert existing CAD drawings in their native format (dwg/dxf) into the file geodatabase. This section will outline the requirements for data conversion from native format to file geodatabase and will be detailed in its respective sub-sections.

3.4.1 Workflow

An Extract, Transform, and Load (ETL) tool will be designed to automate the process of converting CAD drawings stored in native format (dwg/dxf) into GIS data format and migrating them into a centralized geodatabase.

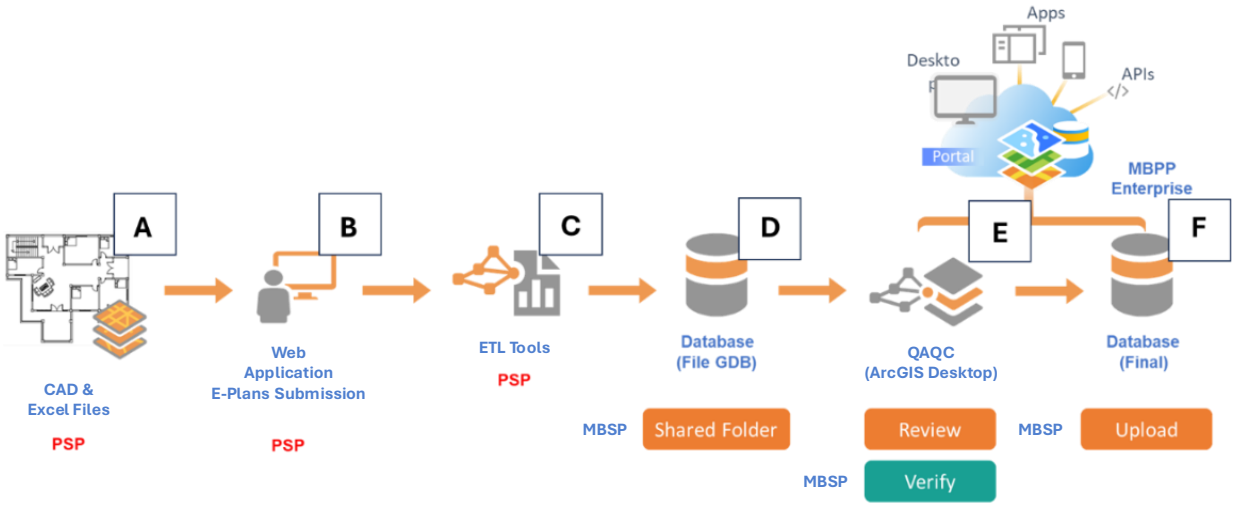


Figure 26: Overall Workflow of CAD File Conversion

Figure above shows the conversion process for CAD & Excel files. Here is the description of each workflow process:

a. PSP will prepare all plans approved at the current OSC+ system level, ensuring compliance with specifications based on CAD & Excel template guidelines.

b. PSP will log into the E-Plan Submission system to upload CAD and Excel files that contain details related to the CAD file.

c. The CAD and Excel files will be converted into GIS format data using an ETL tool, re-project to MY_G_Cassini_Pulau_Pinang_&_Seberang_Prai and structured based on the data model designed. Only layers with a distinguished and recognizable name will be converted using the ETL tool. Any features that are drawn in the unidentified layer's name will not be converted.

d. The converted GIS data will be loaded into the file geodatabase stored in shared folder.

e. After that, MBSP will access the shared folder to get the file geodatabase and undergo QA/QC process in ArcGIS Desktop. This QA/QC process will be done by MBSP respectively to ensure each converted record are accurate and verified.

f. The verified data will be loaded into the final database.

3.4.2 ETL Process

This section will generally describe the ETL process using ArcGIS Data Interoperability to convert a CAD drawing to GIS data. There are four main stages of process that consists of data extraction, data transformation, and data loading.

Following

Figure 27 is the detailed description of the three stages of the process:

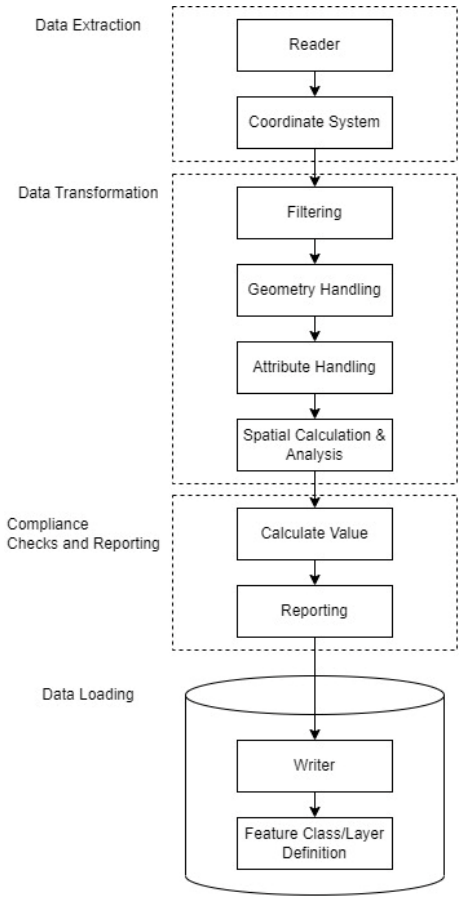


Figure 27: ETL Workflow of CAD Files Conversion

1. Data Extraction

a. Reader: Add a reader specific to your CAD format (e.g., AutoCAD DWG Reader).

b. Coordinate System: Define the coordinate system of your CAD drawing. This is crucial for accurate positioning in the GIS. In this stage, MY_G_Cassini_Pulau_Pinang_&_Seberang_Prai will be used as the main coordinate system.

2. Data Transformation

a. Filtering: (Optional) Use a FeatureTypeFilter or AttributeFilter to select specific layers or features from the CAD drawing for conversion.

b. Geometry Handling:

i. LineCloser: Convert open polylines from CAD into closed polygons for GIS.

ii. Snapper: Snap slightly disconnected lines to create clean geometries.

iii. GeometryValidator: Check for and potentially fix invalid geometries like self-intersections.

c. Attribute Handling:

   i. AttributeManager: Rename, manipulate, or remove unwanted attributes.

   ii. AttributeCreator: Create new attributes based on existing data or calculations.

   iii. AttributeMapper: Map attributes from the CAD drawing to corresponding fields in the GIS format.

d. Spatial Calculations and Analysis:

   i. AreaCalculator/LineLengthCalculator: These transformers will calculate the area or length of the door features, allowing you to compare them to the minimum width requirement. (e.g., minimum door width)

   ii. NetworkBuilder/ShortestPathFinder: These transformers will identify the shortest paths from exits to building egress points. By measuring the path lengths, you can verify compliance with the code. (e.g., exit path lengths)

3. Compliance Checks and Reporting

a. Filtering Based on Rules: After performing spatial calculations, use "MathEvaluator" or "ConditionalFilter" transformers to compare the calculated values (door width, path length) against the pre-defined building code requirements. These transformers will filter out compliant features and identify potential violations.

b. Generating Reports: Utilize "FeatureWriter" or "TextFileWriter" transformers to generate reports highlighting non-compliant features. These reports can include details like feature ID, type and the specific code violation.

4. Data Loading

a. Writer: Choose a writer for your desired GIS format. File Geodatabase Writer will be chosen as a Writer or final output of the conversion process.

b. Feature Class/Layer Definition: Define the name and geometry type (point, line, polygon) for the output features in the GIS format.

3.4.3 QAQC Process

This section will generally describe the QAQC process using the ArcGIS desktop application after conversion process has been done. There are three main stages of process that consists of data accessibility, data verification, and data publishing as show in figure below.

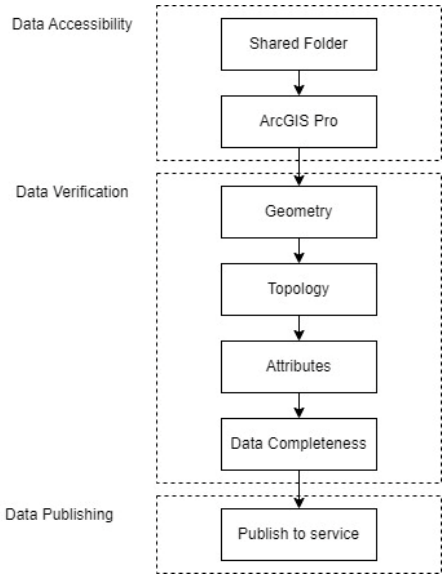


Figure 28: QAQC Workflow using ArcGIS Desktop

1. **Data Accessibility**

a. The output of CAD Conversion which is file geodatabase in E-Plan Submission system will be stored in MBSP's server.

b. To access file geodatabase, need to open ArcGIS Desktop and establish folder connection to the server directory. After successful connection, data verification process can be start.

2. **Data Verification**

a. Geometry: Visually inspect the converted features to ensure they accurately represent the corresponding elements in the CAD drawing.

b. Topology: Validate the topological relationships between features (e.g., do roads connect properly at intersections).

c. Attributes: Verify that attributes have been correctly mapped from the CAD drawing to the feature class table.

d. Data Completeness: Ensure all features and attributes from the CAD drawing are present in the converted feature class.

3. Data Publishing

a. After data verification process done, the finalize data will publish as a service into MBSP Enterprise.

3.4.4 Assumptions

Here are the assumptions on CAD file conversion to file geodatabase:

1. Only CAD layers state in final CAD template will be converted to GIS.
2. Before submission to E-Plan Submission system, the contractors should prepare the CAD files based on the guidelines set by MBSP. MBSP will do a pre-assessment of contractors' CAD files before submission to E-Plan Submission system.
3. The output feature class may not be positioned correctly on the map if the CAD file is not georeferenced based on the correct coordinate system, in which case georeferenced process in ArcGIS Desktop may be required.
4. If the size of the CAD file exceeds the maximum permissible size as mentioned in the Garis Panduan guide, the CAD file will be rejected and shall be submitted by the contractor to MBSP via other means (emails/CD, etc).
5. For CAD file naming convention, there will be a guideline in text form in Garis Panduan document, the system will store submitted files to the server without changing the file names.
6. If required, MBSP's IT Department shall facilitate access to converted CAD/submitted items in MBSP server.
7. The end-to-end workflow for file uploads and data conversion has not been finalized and may be revised during the system development phase.

3.5 Data Model

A data model is an abstract representation that organizes data elements and standardizes their relationships and properties. By utilizing existing industry-standard data models, this solution can provide MBSP with a reliable foundation to systematically and efficiently store GIS datasets in a multi-user environment. This encompasses the storage of geographic features, attribute tables, imagery and analytical models.

3.5.1 Data Structure

Here is a suggestion for a E-Plan Submission data structure that can be used in the implementation of database development. The table below shows the attribute information based on excel template by *Jabatan Perancang Bandar*.

Table 13: The Attribute Information That Will Be Implemented for Each Layer in The Final Geodatabase for Planning Data

NO	FIELD NAME	NAME	DATA TYPE	LENGTH
1	applicationdate	Tarikh Kemuka	Date	
2	no_file	No Permohonan	Text	50
3	tajuk	Cadangan Pemajuan	Text	10000
4	jeniskerja	Jenis Kerja	Text	50
5	namapemohon	Nama Pemohon	Text	50
6	alamatpemohon	Alamat Pemohon	Text	250
7	namapsp	Nama Syarikat	Text	50
8	mjoscid	MJOSC ID	Text	50
9	mjoscdate	Tarikh Mesyuarat	Date	
10	mjoscresult	Status Permohonan	Text	50

11	diprosesoleh	Diproses Oleh	Text	50
12	nokm	No KM	Text	50

3.5.2 Projection System

The MY_G_Cassini_Pulau_Pinang_and_Seberang_Prai coordinate system will serve as the primary reference. Coordinate system reprojection will occur automatically during the ETL process. All projected data will undergo quality assurance and verification processes using ArcGIS Desktop applications.

3.5.3 List of GIS Layers by Color/Symbology, and Feature Geometry

The list of layers by color/symbology and feature geometry for planning GIS data.

Table 14: List of Color/Symbology and Feature Geometry

NO	LAYER NAME	COLOR CODE	FEATURE GEOMETRY
1	Plot	HEX: #0070FF	Polygon
2	Footprint	HEX: #FF00FF	Polygon
3	Road	HEX: #38A800	Line

FUNCTIONAL REQUIREMENTS

4.1 Details Requirement

Table 15: Detail Requirements

NO	USER	REQUIREMENT	ACCEPTANCE CRITERIA	REFERENCE
1	Super Admin	Admin Listing	Super Admin can view a list of all registered Super Admin and Admin.	
2	Super Admin / Admin	PSP User Listing	Super Admin or Admin can view a list of all registered PSP users.	
3	All Users	Profile Setting	All users can update their own profile.	
4	All Users	Profile Setting	All users can update their password.	
5	PSP Users	Plan Submission Management	PSP users can view a list of only their own submitted items. The list will display items with the following statuses: • *Draf* (Draft) • *Menunggu Pengesahan* (Awaiting approval) • *Diterima* (Verified)	

NO	USER	REQUIREMENT	ACCEPTANCE CRITERIA	REFERENCE
			• *Ditutup* (Discontinued) • *Mohon Pembetulan* (Request revision)	
6	PSP Users	Plan Submission Management	PSP users can perform an initial submission by uploading the development plan, including CAD drawing files and excel, along with the required application detail.	
7	PSP Users	Plan Submission Management	PSP users can review a summary of their plan submission prior to finalizing and sending it to the Admin for approval.	
8	Super Admin / Admin	Plan Approval Management	Super Admins and Admins can view a list of all plans applications submitted by every PSP user. The list will display items with the following statuses: • *Menunggu Pengesahan* (Awaiting approval) • *Diterima* (Verified) • *Ditutup* (Discontinued) • *Mohon Pembetulan* (Request revision)	
9	Super Admin / Admin	Plan Approval Management	Super Admins and Admins can update the status of submissions to one of the following statuses: • *Diterima* (Verified)	

NO	USER	REQUIREMENT	ACCEPTANCE CRITERIA	REFERENCE
			• *Ditutup* (Discontinued) • *Mohon Pembetulan* (Request Revision)	
12	PSP Users	CAD File Conversion to GIS	Allow PSP Users to submit planning permission plan in DWG/DXF format.	
13	PSP Users	CAD File Conversion to GIS	Allow PSP Users to submit excel file.	

4.2 List of Data

Table 16: List of Data

DEPARTMENT	DATA SOURCE	STATUS	REMARK
Jabatan Perancang Bandar	Sample CAD Drawing	Complete	Contractor received in DWG format.
Jabatan Perancang Bandar	Excel Template	Complete	Contractor received in .xlsx format.
Jabatan Perancang Bandar	Existing GIS data for permohonan kebenaran merancang.	Pending	MBSP needs to provide existing GIS data for permohonan kebenaran merancang in file geodatabase format.

CONCLUSION

This document is prepared based on information received from the User Requirements Study session that was conducted and mutually agreed upon between the project team of the Majlis Bandaraya Seberang Perai (MBSP) and Esri Malaysia Sdn Bhd. This document will also serve as an action plan or blueprint that will be used as a reference throughout the development period of this project.